Exam Date & Time: 02-Dec-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

DIGITAL SYSTEM DESIGN [BME 2123]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

1A) Design an 8:1 MUX using 2:1 MUX. Provide a block diagram and a truth table to illustrate the design. (4)

1B) Design a full subtractor, including a) a block diagram, b) a truth table, c) a Boolean expression, and d) a logic diagram (4)

1C) Compare and analyse two differences between combinational and sequential circuits. (2)

2A) Implement a 2:1 MUX for the given function
f = w1.w2 + w1.w3 + w2.w3
by applying the Shannon expansion theorem, without utilizing basic gates (4)

2B) Analyse and deduce the given Boolean function using the CMOS transmission gate
$F = AB + \overline{A}.\overline{C} + A.\overline{B}.C$ (4)

2C) Explain and demonstrate Shannon's expansion theorem using a block diagram and relevant equations (2)

3A) Analyze and implement the following function using Programmable Array Logic (PAL)
$f1 = x_1 x_2 x_3 + x_1 x_2 x_3$ and $f2 = x_1 x_2 + x_1 x_2 x_3$ (4)

3B) Analyze and implement the following function using a Programmable Logic Array (PLA)

$f1 = x_1 x_2 + x_1 \overline{x_3} + \overline{x_1}\,\overline{x_2} x_3$ (4)

$f2 = x_1 x_2 + \overline{x_1}\,\overline{x_2} x_3 + x_1 x_3$

3C) Compare and illustrate two differences between Programmable Array Logic (PAL) and Programmable Logic Array (PLA). (2)

4A) Analyze and illustrate the operation of a Master-Slave J-K Flip-Flop using NAND gates and its corresponding timing diagram (4)

4B) Analyze and implement the minimum CMOS transistor network that realizes the (4)

functionality of the Boolean equation
Y = A . (B + C) + C.D

4C)	Explain Moore's law with the corresponding diagram	(2)
5A)	Analyze and design a CMOS SR Latch using NAND gates, including a) the circuit diagram, b) the truth table, and c) the pull-up and pull-down circuits	(4)
5B)	Analyze and design a two-input LUT for the given function $f = x_1 . x_2 + x_1 x_2$	(3)
5C)	Discuss 4-bit Johnson's counter with the help of D flip flop and truth table	(3)

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